# JAFFEDRONE

**A drone instrument and meditation tool for the TrimUI Brick and Brick Pro**

User manual · version 0.4.2




github.com/sydilaxe/Trimui-Jaffedrone-dev

# Getting started




Jaffedrone is a drone of five voices: strings that are bowed and plucked, or wind voices that are blown and tongued, tuned to a scale and moving through chords. Each voice has its own rhythm, or you play it by hand.

## Install

Unzip Jaffedrone-brick.zip (TrimUI Brick and Brick Hammer) or Jaffedrone-brickpro.zip (Brick Pro: it adds the sticks) at the root of the SD card (it holds Apps/Jaffedrone/) and open Apps → Jaffedrone. The start-up screen shows for a moment; any button moves on.

## The first buttons

| | | |
|---|---|---|
| 1 | START | start or stop the drone (it fades in and out) |
| 2 | START + L1 / R1 | previous / next page: PLAY, SOUND, COMPOSE, MIX, SONG |
| | SELECT | the next chord |
| 3 | MENU | leave the app |

Listen on headphones for the deepest low end and for binaural beats; the speaker plays little below about 200 Hz.

## The five pages



PLAY: the strings and their rhythms



SOUND: each voice's tone



COMPOSE: scale, chords, change



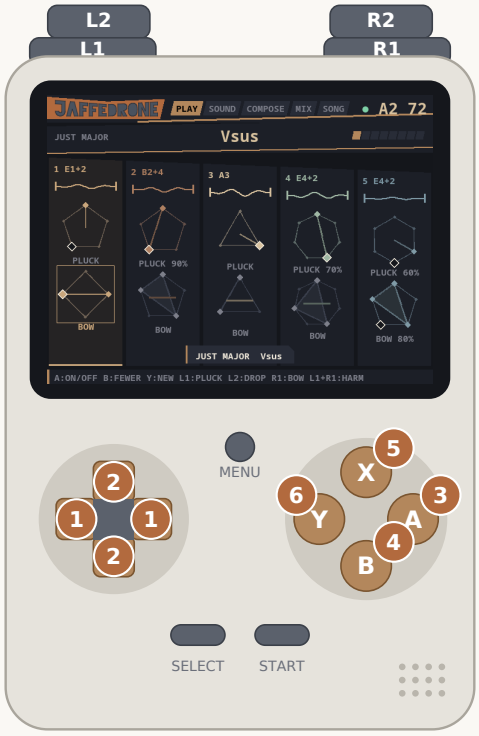
MIX: levels and effects



SONG: presets, chord memory, key

START + R1 moves right along the tabs at the top, START + L1 left. On SOUND, COMPOSE and MIX, L1 and R1 alone change page too.

# PLAY: the rhythm rings

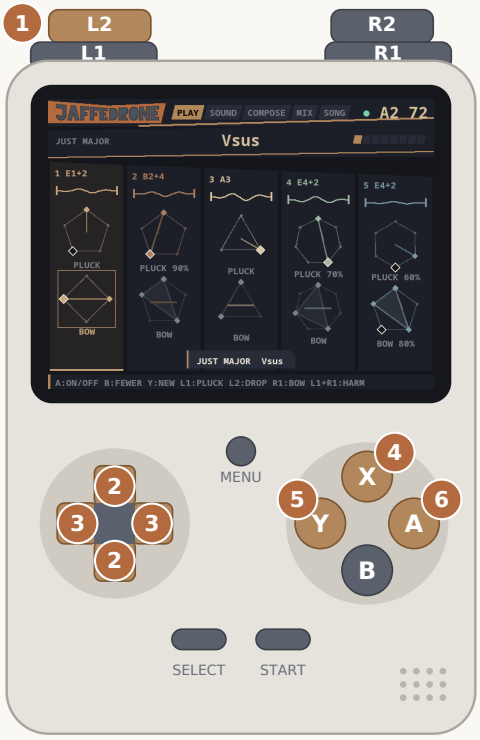



Each string has two Euclidean rings: the pluck ring plucks on its pulses and the bow ring swells the bow on its pulses (off, the bow plays steadily). A ring spreads its pulses as evenly as it can over its steps, like the rhythms of HI5 and RadioScan.

| | | |
|---|---|---|
| 1 | ◄► | choose a string |
| 2 | ▲▼ | choose its pluck ring or bow ring |
| 3 | A | switch the ring on or off |
| 4 | B | fewer pulses, wrapping back to all |
| 5 | X | rotate the pattern |
| 6 | Y | a new simple pattern |

A label like PLUCK 70% means each pulse of that ring plays seven times in ten (set with R2 + X / Y).

## Shaping a ring: hold L2




| | | |
|---|---|---|
| 1 | hold L2 | then: |
| 2 | ▲▼ | more / fewer steps |
| 3 | ◄► | step length (right is faster): 1/16 to 2 bars |
| 4 | X | more pulses |
| 5 | Y | rotate back |
| 6 | A | mute or unmute the string (a red bar marks it) |

# Playing by hand

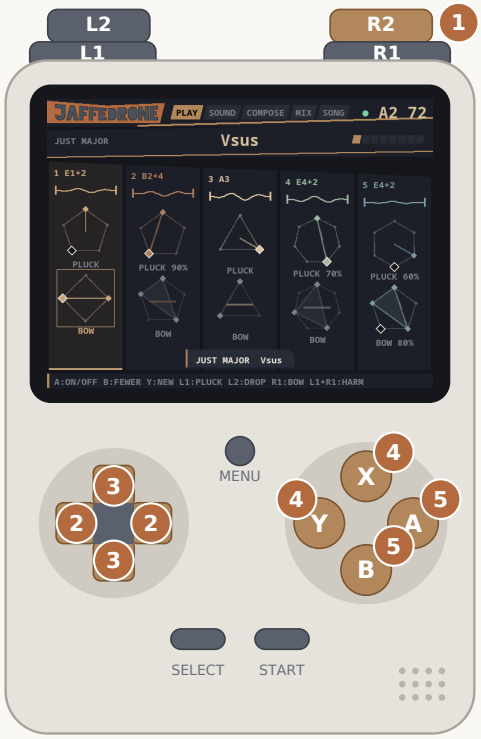
The shoulder buttons play the string selected on PLAY, over whatever its rings are doing. A wind voice is tongued and blown instead.




| | | |
|---|---|---|
| 1 | L1 | pluck the selected string, the moment you press (wind: tongue a note; struck: the mallet) |
| 2 | L2 | drop a dulcimer mallet on it: a bright hit, then the mallet bounces, each rebound lower and sooner, until it rests (wind: a flutter tongue, the breath rolling "frrr" and settling; struck: a hard mallet, bouncing). Held, L2 still shapes the ring |
| 3 | R1 held | bow it: the bow is drawn in slowly, swelling over about two seconds and brightening as it bites, and sings while you hold (wind: blow) |
| | L1 + R1, either first | bow with a light touch at a node: the next harmonic up sings (wind: overblow, an octave up, a twelfth for the reed). Bowing with R1, add L1 for overblows as you bow |

The bowed string's column lights while the bow is on it.

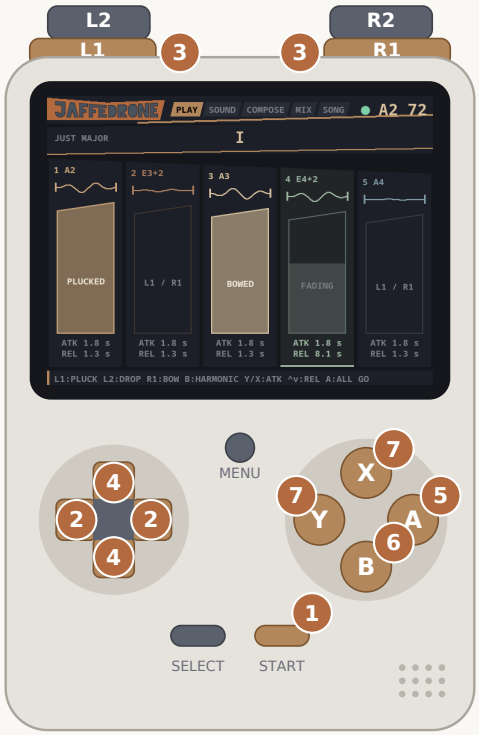
## Quick sound changes: hold R2




| | | |
|---|---|---|
| 1 | hold R2 | then: |
| 2 | ◄ ► | previous / next scale |
| 3 | ▲ ▼ | tone: brighter / darker |
| 4 | X / Y | the selected ring's chance, more / less |
| 5 | A / B | body resonance, more / less |

# LATCH: hold each string yourself

In LATCH the rings rest and every string is silent until you play it. A string you pluck or bow keeps droning, with no need to hold the button, until you play it again; then it fades over its own RELEASE.




| | | |
|---|---|---|
| **1** | START + A | switch PLAY MODE between RINGS and LATCH (also on the SONG page) |
| **2** | ◄ ► | choose a string |
| **3** | L1 / R1 | pluck / bow it on: it drones (L2: the mallet drops it on). Again: it lets go and fades |
| **4** | ▲ ▼ | that string's RELEASE, 0.05 s to 20 s |
| **5** | A | let every string go |
| **6** | B held, or L1 + R1 | the next harmonic up on the droning string (wind: overblow); with both shoulders, either first, it lasts while both are held |
| **7** | Y / X | that string's bow ATTACK, shorter / longer, 0.05 s to 10 s |

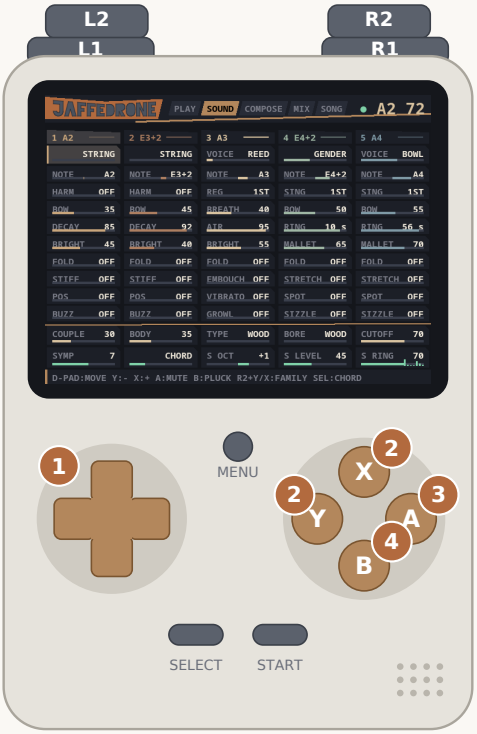
Each column shows a board with a shed-roof top. It fills while its string drones and drains as it fades, marked STRUCK, PLUCKED, BOWED, HARMONIC or FADING, with its ATTACK and RELEASE beneath. A pluck or strike starts at once; a bow swells in over its ATTACK. RELEASE is also on the MIX page; LATCH, ATTACK and RELEASE are saved with songs and presets.

## Try it

Press START + A. Bow string 1 with R1 for a low drone, move to string 3 and pluck it with L1, then press L1 on string 3 again to let it ring away. Raise string 5's RELEASE with ▲ to ten seconds or more for a long dying shimmer. Tap L2 on string 4 to drop the mallet on it, and hold B to let its harmonic sing. Lower string 1's ATTACK with Y for a bow that bites at once.

# SOUND, COMPOSE and MIX

On these pages the d-pad moves to the nearest control in that direction, and Y / X lower / raise it. Held, they speed up.




| | | |
|---|---|---|
| 1 | d-pad | move between controls |
| 2 | Y / X | lower / raise the control |
| 3 | A | on a string's control: mute or unmute that string |
| 4 | B | pluck the string (to hear a change) |

## SOUND

The VOICE row makes each column a STRING, one of seven winds (LIP, REED, DOUBLE, FLUTE, FREE, RECORDER, BANSURI) or one of nine struck voices (see Struck voices). R2 + Y / X on VOICE jumps to the previous / next family; Y / X alone step one at a time. The rows below follow the voice: a string has BOW, DECAY, BRIGHT, FOLD, STIFF, POS and BUZZ; a wind BREATH, AIR, EMBOUCH, VIBRATO and GROWL; a struck voice BOW, RING, MALLET, STRETCH, SPOT, SIZZLE and SING. The shared row holds COUPLE, BODY, its TYPE, the winds' BORE and CUTOFF; the last row the sympathetic strings.




COMPOSE: the scale and root (a wheel of the scale's notes, the chord lit), how often chords change and how: MOTION, HOME, COLOR, CHANCE, RELATE, EVOLVE, GLIDE, tempo, Wi-Fi sync and two LFOs.




MIX: each voice's LEVEL, PAN, GATE (RELEASE in LATCH) and AURA, the master, undertones (SUB), then AURA, delay, reverb and binaural.

# AURA: voices that listen

AURA lets some voices sound only from the others, as if they stood beside them in one room: a gong humming along with a flute, bowls singing when a reed plays. On MIX, set each voice's AURA to SEND (it plays, and its sound goes into the air), LISTEN (it is never bowed, blown, struck or plucked; it sounds only from what reaches it) or OFF (it plays, out of the air). Then raise AMOUNT under AURA, from OFF to 100.

Each voice stands where its PAN puts it: the players' sound reaches a listener later and fainter the further it stands. A listener rings where its own notes meet the players': struck bodies with a long RING sing most. The sound only goes one way, from players to listeners (and from one listener to the next), so it can never feed back. With every voice on SEND, AURA does nothing: at least one must listen. Delay, reverb and binaural come after, on everything.

# Struck voices

A struck voice is a physical model of a vibrating body: its resonant modes, struck by a mallet (a mass on a spring that meets the surface and bounces off, so a hard mallet makes a short bright contact and a soft one a long dark push) or bowed (stick-slip friction, so the modes it meets grow until they sing, as on a rubbed bowl or a bowed cymbal). Every voice can be struck and bowed.

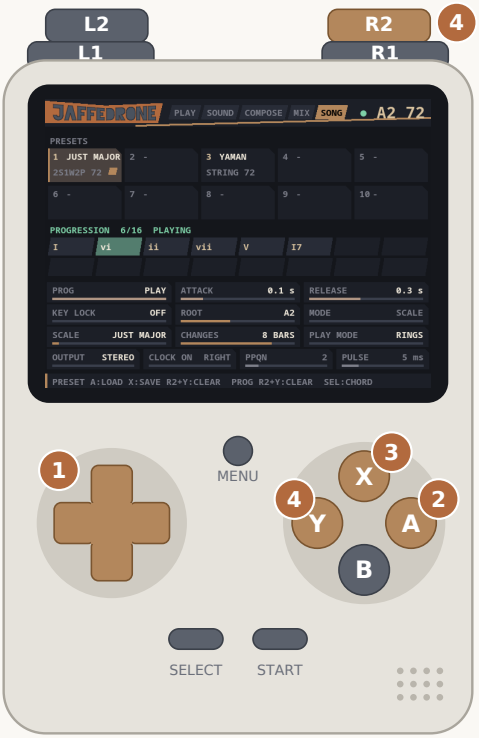



| | | |
|---|---|---|
| **1** | L1 | strike (L2: a hard mallet, dropped, bouncing) |
| **2** | R1 | bow: the body catches and sings |
| **3** | L1 + R1, B | a touch: the fundamental is damped and the next mode rings and sings |

| | |
|---|---|
| **GENDER** | gamelan metallophone bars, in beating pairs (ombak) |
| **BONANG** | gamelan kettle gongs, paired |
| **GONG** | a large gong, low, its pairs beating slowly |
| **BOWL** | a singing bowl: struck, or rubbed until it sings |
| **HANDPAN** | fundamental, octave and fifth above, tuned |
| **STEELPAN** | the same family, brighter |
| **TONGUE** | a steel tongue drum: pure, a bar held at one end |
| **CYMBAL** | a dense shimmer of modes; bowed, it screams and sings |
| **DRUM** | a membrane: a short pitched thud |
| **GLASS** | a glass harmonica: pure, long, made to be bowed |
| **CHORD** | metal tuned to the scale: it rings as a chord |
| **ALIEN** | golden-ratio modes: a metal no body has |
| **RING** | how long it rings; from 95 up a metal drone (DRONE) |
| **MALLET** | soft felt to hard metal |
| **STRETCH** | pulls the modes apart, far at the top: unearthly clangor |
| **SPOT** | where it is struck and bowed: which modes speak |
| **SIZZLE** | a rattle that follows the ring |
| **SING** | the mode the bow favours |
| **BOW** | bow pressure: from a whisper to a scream |

Each struck body has its own limiter, so strikes on a long ring never pile up. The mode frequencies follow measurements and theory: rectangular-bar modes and the makers' harmonic partials for the gamelan bars (Sethares), 1 : 2 : 3 for handpan and pan, 1, 2.9, 5.7, 9.1 for the bowl (Inacio, Henrique and Antunes), a clamped bar for the tongue drum, a thin plate's dense series for the cymbal, the Bessel-function modes of a membrane for the drum.

# SONG: presets, chord memory and key




| | | |
|---|---|---|
| 1 | d-pad | choose a preset slot or a setting |
| 2 | A | load the preset (your Wi-Fi and output settings stay) |
| 3 | X | save the sound and rhythms to the slot |
| 4 | R2 + Y | clear the slot, or the chord memory |

Ten preset slots keep everything: voices, rhythms, scale, effects, the chord memory and PLAY MODE.

## Chord memory

Set PROG to RECORD and every chord change (SELECT or CHANGES) is kept, up to 16. PLAY steps through them round and round; FREE lets the algorithm choose again.

## Playing with others

KEY LOCK holds the root, scale and MODE (the mode chords come from in EQUAL) and tunes chords exactly. ATTACK and RELEASE set how the drone fades in and out with START.

## Output: mono + clock

OUTPUT MONO+CLK plays the sound in mono on one side of the headphone jack and clock pulses on the other, to lead a Pocket Operator, Volca or other gear with a sync input (use a stereo-to-two-mono cable). CLOCK ON picks the side, PPQN the pulses per quarter note (2 for Pocket Operators and Volcas, 24 for DIN sync) and PULSE the width. The pulses run while the drone plays, on the beat and with Wi-Fi sync. Every PAN and stereo spread plays from the centre; the settings are kept for STEREO, and binaural becomes OMBAK.

# The quiet view




START + L2 on PLAY for the quiet view: the five strings across the screen with their rings, and only the chord in words. Again to come back.

| | | |
|---|---|---|
| 1 | START + L2 | the quiet view on / off |

# Binaural, ombak and tunings

On MIX, BEAT sets a binaural beat from 0.1 to 40 Hz: the voices below about 1 kHz are shifted down in the left ear and up in the right, so every low partial beats between the ears. DEPTH sets how much. MODE:

| | |
|---|---|
| **FREE** | the beat as set |
| **TEMPO** | the beat moves to a power-of-two multiple of the tempo |
| **A>T 10M** | a journey from alpha (10 Hz) down to theta (6 Hz) over ten minutes |
| **T>D 15M** | from theta (6 Hz) down to delta (2.5 Hz) over fifteen minutes |
| **OMBAK** | the beat for both ears, like a Balinese gamelan's paired instruments: the low end swells and fades, and it works on the speaker |

## Settings to start from

| | |
|---|---|
| **Meditation** | BEAT 6 MEDITATE, headphones; or MODE A>T 10M to drift down from alpha |
| **Sleep** | MODE T>D 15M, headphones, a long ATTACK and RELEASE |
| **Gamelan shimmer** | scale TIRUS or BEGBEG, MODE OMBAK, BEAT 7 Hz, on the speaker |

The research on binaural beats is limited and mixed; these are the settings most used, not a promise of an effect.

## Tunings

COMPOSE, or R2 + ◄ / ► on PLAY, chooses the scale: EQUAL (with its modes), just major and minor, the harmonic series, just pentatonics, the ragas YAMAN, BHAIRAV, KAFI, BHUPALI and MALKAUNS, the gamelan tunings SLENDRO, PELOG and SELISIR, and the Balinese gong kebyar tunings BEGBEG, SEDANG and TIRUS (Toth's profiles, as reproduced by Vitale and Sethares). Each scale has its own way of moving through chords.

# USB: MIDI and audio

The Brick's USB-C port takes class-compliant MIDI and audio devices (through an OTG adapter or a hub): keyboards, controllers, sequencers, drum machines and USB sound cards. They are found as they are plugged in; a message names each one.

## MIDI in

| | |
|---|---|
| **Notes** | each key takes a voice and tunes it to the key played, on top of the song (the song is never changed). The first key plays the voice selected on PLAY; each key added while others are held takes the next free voice, so chords build across the voices and overlapping keys rotate through them; with all five held, the key held longest gives up its voice. Muted voices are passed over. A key plucks as hard as it is struck and bows while held. A wind voice is tongued and blown. In LATCH a key starts the string droning (a soft key swells in with the bow, a firm one plucks) and letting go fades it over its RELEASE |
| **Pitch bend** | two semitones |
| **Mod wheel, pressure** | how hard the held strings are bowed |
| **CC64** | sustain |
| **CC74** | TONE: darker / brighter |
| **CC91** | more / less reverb |
| **Clock** | the tempo follows it; START, CONTINUE, STOP and song position start, stop and place the drone |

## MIDI out

While the drone plays, and no clock is coming in, Jaffedrone sends MIDI clock (24 a quarter note) with START and STOP to every MIDI port, timed to the sound as it is heard. It leads a drum machine or sequencer over USB as MONO+CLK does over the headphone jack.

## Audio

A USB sound card or DAC, once plugged in, takes the sound from the speaker and headphones; unplugged, the sound comes back to them.

## Settings

In Apps/Jaffedrone/launch.sh: JD_MIDI_NOTES=CHANNELS lets MIDI channels 1-5 play strings 1-5; JD_MIDI_CLOCK=0 sends no clock; JD_MIDI_FOLLOW=0 ignores the clock coming in; JD_AUDIO=internal keeps the sound on the speaker and headphones.

# Brick Pro: the sticks




On the Brick Pro the sticks play the string selected on PLAY, and spring back when let go: nothing they do is saved. Each stick reaches its controls' whole range, straight or on a diagonal: it learns how far it really goes the first time it is pushed all the way.

| | | |
|---|---|---|
| 1 | left stick ▲ | bow the string by hand: a light touch draws the bow lightly, all the way up as hard as the bow goes, half again as hard as R1 (a wind voice: blow) |
| | left stick ◀ ▶ | bend the string, up to two semitones either way |
| 2 | right stick ▲ ▼ | HARM: up touches higher harmonics, to the 5th; down, lower, to none (a wind voice: its register; struck: the mode the bow sings) |
| | right stick ◀ ▶ | the SOUND control on the stick (BRIGHT at first): right up to its most, left down to its least |
| | L3 (left click) | pluck the string |
| | R3 (right click) | the next chord; on SOUND, put the row under the cursor on the right stick |

JD_BEND=7 in launch.sh bends up to seven semitones.

## Choosing the right stick's control



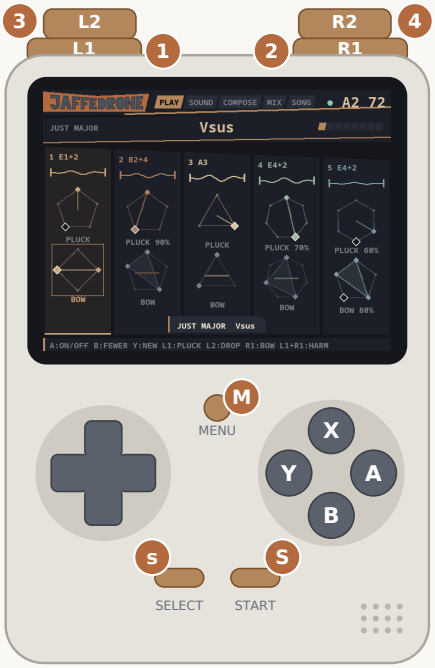

On SOUND, move to a row (BOW, DECAY, BRIGHT, FOLD, STIFF, POS, BUZZ or NOTE) and click the right stick 1: its left / right now plays that control. The choice is kept. The rows the stick plays are marked on the selected string's column, and light up while the stick moves them; the row shows the value playing.

## The lights

The lights around the sticks show how far they are pushed: the left glows amber as it bows and rose as it bends, the right green for the harmonics and gold for its SOUND control, brighter the further each goes. At rest they are dark; on leaving, they show what they showed before.

In launch.sh: JD_LEDS=0 leaves the lights alone; JD_STICK_REACH=80 takes a stick to reach 80% of the way before it shows how far it goes (60 at first).

# Quick reference




| | |
|---|---|
| START | start / stop the drone |
| START + L1/R1 | previous / next page |
| START + A | PLAY MODE: RINGS / LATCH |
| **LATCH** B / YX | harmonic (held) / bow attack |
| START + SELECT | the log: OFF / BASIC / DETAIL |
| SELECT | next chord |
| L1 | pluck (wind: tongue); LATCH: on / let go |
| L2 | drop the dulcimer mallet (wind: flutter tongue) |
| R1 | bow while held (wind: blow); LATCH: bow on / let go |
| L1 + R1 | harmonic (wind: overblow), either first |
| START + L2 | quiet view |
| L2 + ... | shape the ring, mute (A) |
| R2 + ... | scale, tone, chance, body |
| MENU | leave |

## The log

START + SELECT cycles the log through OFF, BASIC and DETAIL. It is written to Apps/Jaffedrone/jaffedrone_log.txt: how hard the audio is working, any dropouts, the levels, each voice's tuning (DETAIL) and what was played, with suggestions when something could be better. If you hear crackles, set JD_BUFFER=1024 in Apps/Jaffedrone/launch.sh.